

Week 10: Modelling Asymmetry in Volatility

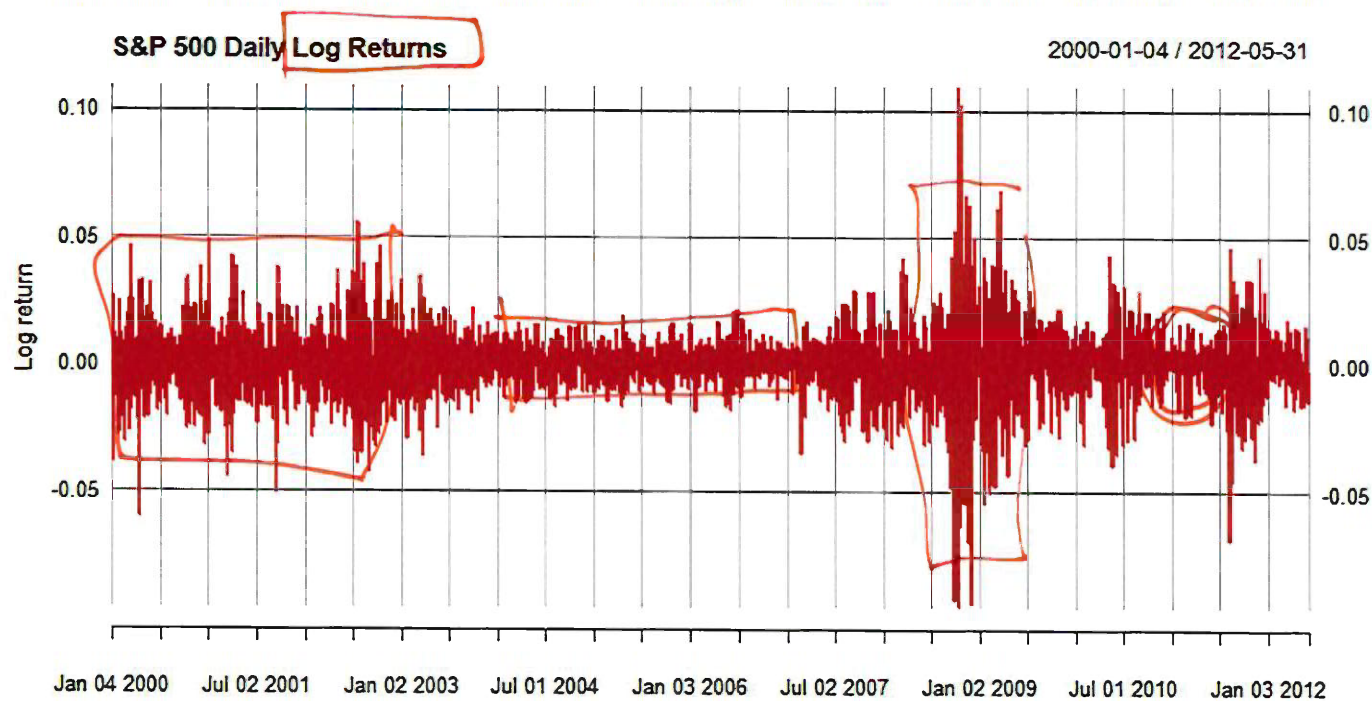
Consider S&P 500 daily series: January 2000 - May 2012.



← greater
volatility
"panic" →

smaller
volatility.

← →

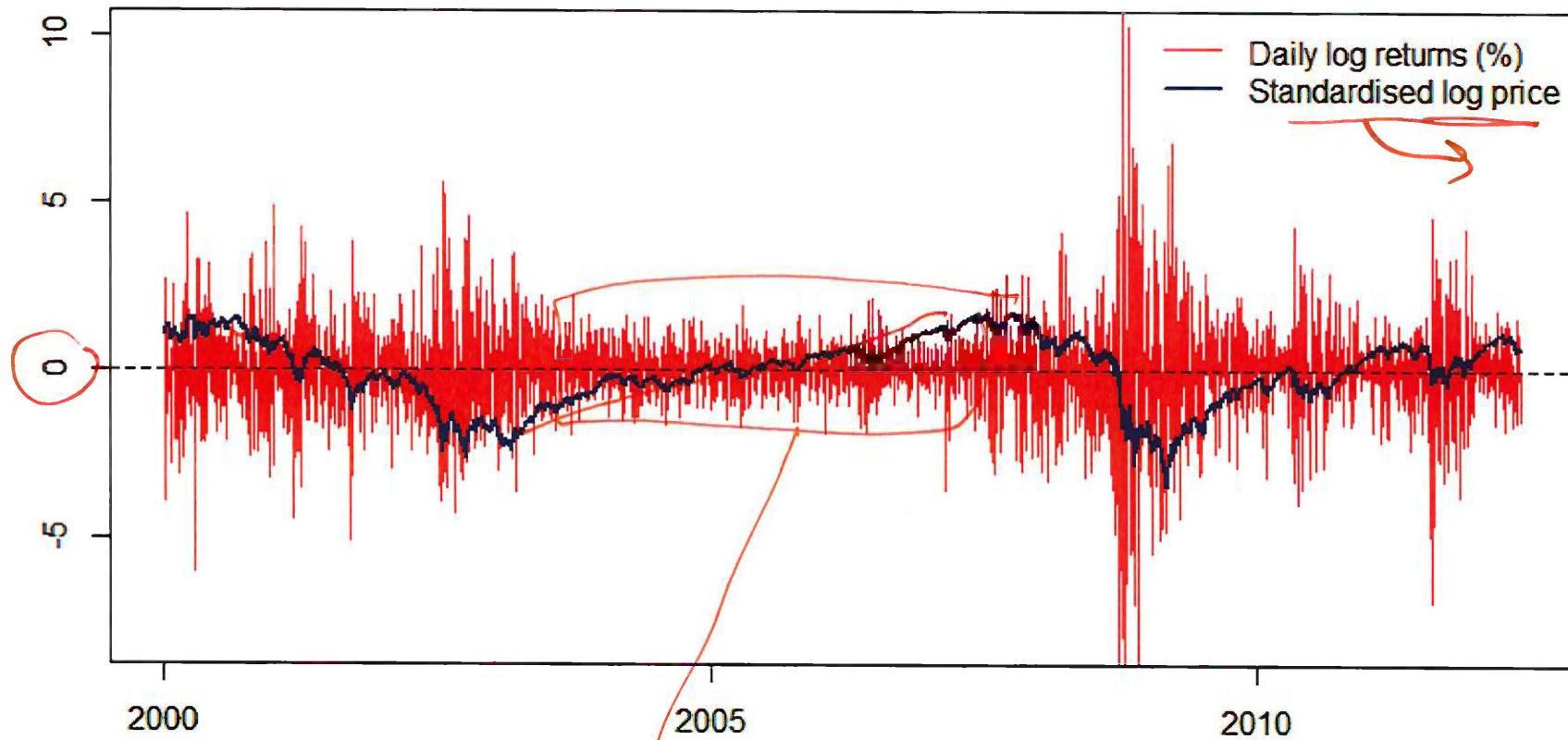


For fun.

S&P500 Daily returns and prices

① log

② z



$$z = \frac{x - \mu}{\sigma}$$

Falling prices are often associated with higher volatility...

All Ordinaries index

Consider the daily stock prices on the Australian All Ordinaries index from 24 Feb 2003 to 30 September 2019.

Stock prices trend upwards and then crash during the GFC, 2008-09. By the end of the sample, prices had only recently returned to their previous peak.

The plot of stock log returns suggests two regimes:

- Tranquillity: a period of calmness where volatility is relatively low and remains low.
- Turbulence: a period of large movements in (log) prices such as during the GFC, which persists over time where volatility is high and stays high. panic

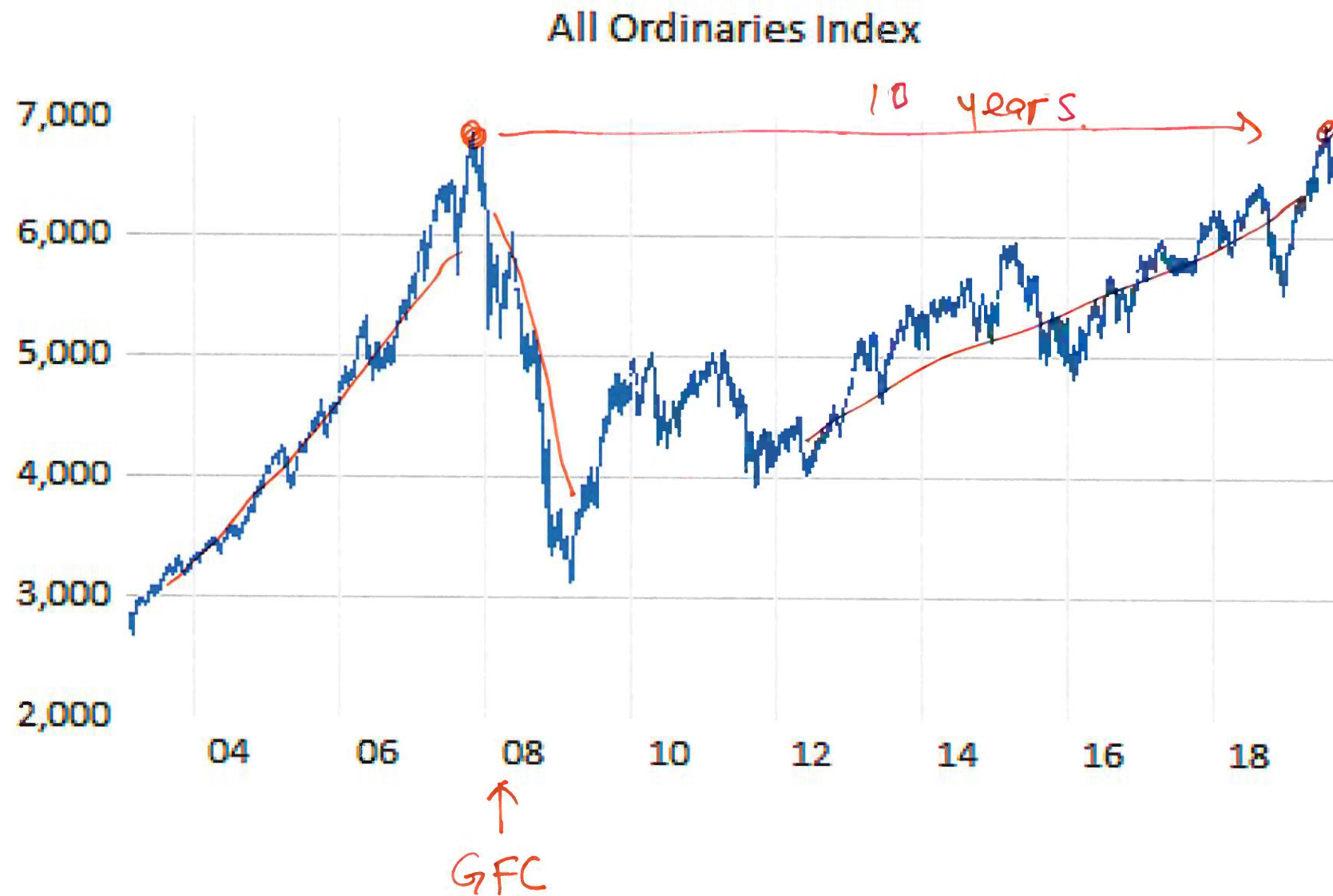
```
> df <- readxl::read_excel("all_ord.xlsx")
>
> library(xts)
> data.xts <- as.xts(df)
> tail(data.xts)
```

	All_ord
2019-09-23	6861.1
2019-09-24	6856.6
2019-09-25	6814.7
2019-09-26	6785.6
2019-09-27	6824.1
2019-09-30	6800.6

```
> log_ret <- na.omit(diff(log(data.xts$All_ord))*100)
> colnames(log_ret) <- "log return"
> tail(log_ret)
```

	log return
2019-09-23	0.32262709
2019-09-24	-0.06560867
2019-09-25	-0.61296341
2019-09-26	-0.42793381
2019-09-27	0.56577444
2019-09-30	-0.34496206

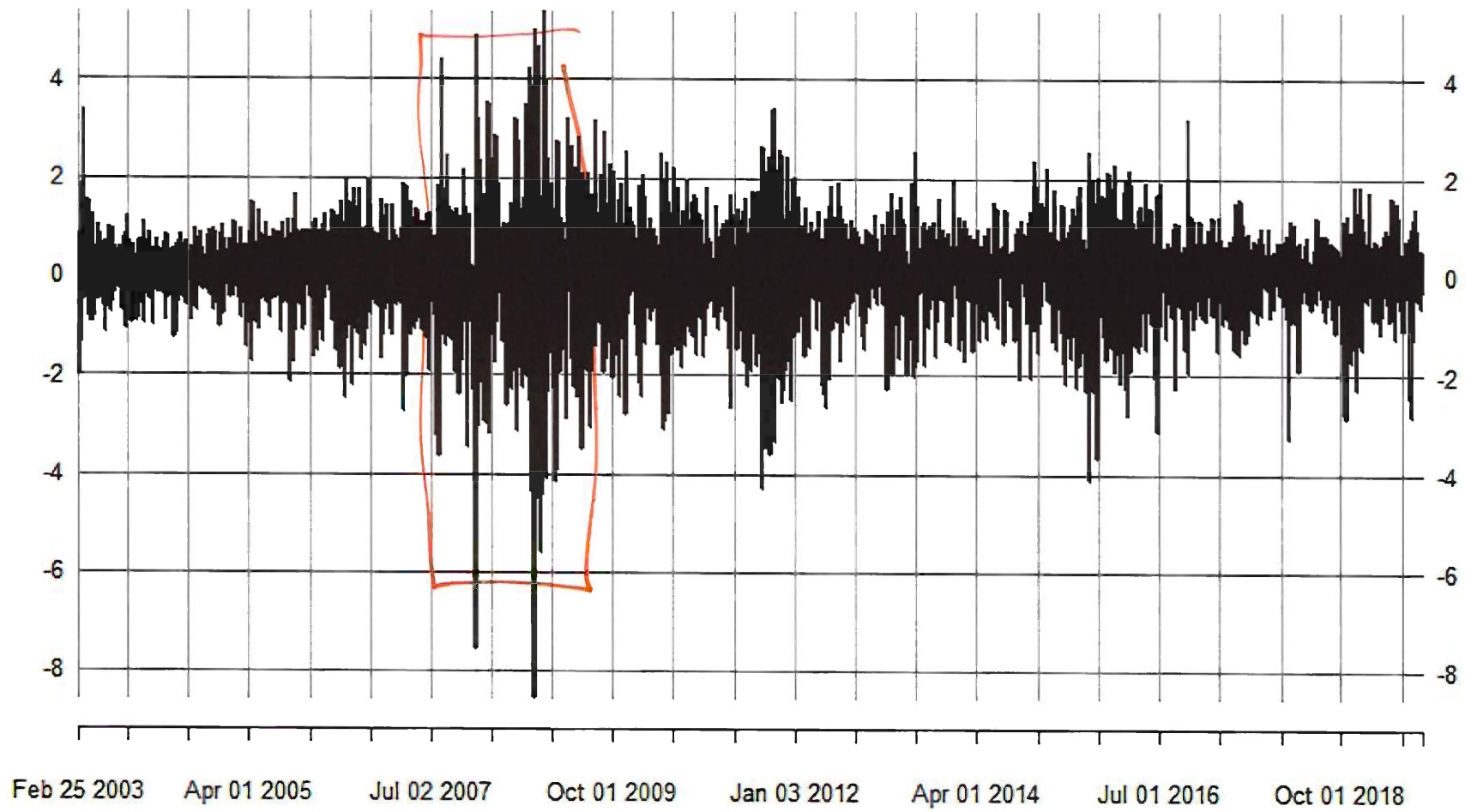
%



log_ret

GFC.

2003-02-25 / 2019-09-30



GARCH(1,1)

①

```
spec_garch11 <- ugarchspec(  
  variance.model = list(  
    model = "SGARCH",           # standard GARCH  
    garchorder = c(1, 1)        # c(ARCH, GARCH)  
  ),  
  mean.model = list(  
    armaorder = c(0, 0),        # no AR term.  
    include.mean = TRUE         # just an intercept (constant mean)  
  ),  
  distribution.model = "norm"   # normal errors  
)
```

§ ②

```
# Estimate the models  
fit_garch11 <- ugarchfit(  
  spec = spec_garch11,  
  data = log_ret  
)  
      ↓  
    data.
```

GARCH(1,1)

$$y_t = \hat{\mu} + \varepsilon_t$$

```
> fit_garch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.054277155	0.010361249	5.238476	1.619079e-07
omega	0.007831524	0.001842924	4.249510	2.142384e-05
alpha1	0.092056042	0.009446833	9.744646	0.000000e+00
beta1	0.900680594	0.009854408	91.398753	0.000000e+00

wow

^
α/0

GARCH(1,1) model

$$r_t = c + \varepsilon_t$$

$$\sigma_t^2 = \underbrace{\alpha_0}_{>0} + \underbrace{\alpha_1}_{\geq 0} \varepsilon_{t-1}^2 + \underbrace{\beta_1}_{\geq 0} \sigma_{t-1}^2$$

The fitted model

$$r_t = 0.054 + \hat{\varepsilon}_t$$

$$\hat{\sigma}_t^2 = \overset{\checkmark}{0.0078} + \overset{\checkmark}{0.092} \hat{\varepsilon}_{t-1}^2 + \overset{\text{wow}}{0.901} \hat{\sigma}_{t-1}^2$$

1. All the estimated coefficients (i.e., $\hat{\alpha}_0$, $\hat{\alpha}_1$, and $\hat{\beta}_1$) are positive. ✓
2. $\hat{\alpha}_1 + \hat{\beta}_1 = 0.092 + 0.901 < 1$ and so $\mathbb{V}(\varepsilon_t) > 0$ and hence the GARCH(1,1) model has a finite unconditional variance.

$\frac{1}{0}$ = not finite $v(\varepsilon_t)$ exist.

Leverage effects

Leverage effects in conditional variance models refer to the phenomenon where **negative shocks** (bad news) to asset returns tend to **increase future volatility** more than **positive shocks** (good news) of the same magnitude.

Fischer Black notes that this **asymmetric** response of volatility to shocks is observed in many financial time series and reflects the idea that markets react more strongly to negative information than to positive.

Why it is called the leverage effect?

$$\uparrow \text{ratio} = \frac{\text{debt} = \text{fixed}}{\text{equity} \downarrow} = \text{higher risk.}$$

The term originates from the financial leverage of firms: when a firm's stock price drops (a negative shock), its **debt-to-equity ratio rises**, increasing its perceived risk and hence its volatility.

When bad news causes a firm's stock price to fall, its market **equity** (= price $\downarrow$ $\times$ number of shares) value decreases, while its debt remains unchanged (at least in the short term).

This automatically causes the debt-to-**equity** ratio to rise, because the denominator (**equity**) is now smaller.

Why does this matter? **Higher leverage** means **higher financial risk**. $\Rightarrow$ **higher volatility**
 $\uparrow$ ratio

Investors and analysts perceive the firm as riskier, leading to higher expected future volatility.

Asymmetric Volatility Models

We will consider two asymmetric volatility models:

1. GJR-GARCH (Threshold GARCH)
2. EGARCH.

Asymmetric Volatility Models

A threshold GARCH model, due to **G**losten, **J**agannathan and **R**unkle is **GJR**-GARCH:

$$r_t = \mu + \varepsilon_t$$

$$\varepsilon_t = \sigma_t u_t$$

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \overbrace{\lambda(I_{t-1} \varepsilon_{t-1}^2)}^{\text{new.}} + \beta_1 \sigma_{t-1}^2$$

$$I_{t-1} = \begin{cases} 1 & \varepsilon_{t-1} \leq 0 \\ 0 & \text{else} \end{cases}$$

Focus

bad news $\Rightarrow$ negative shock.

good news $\Rightarrow$ prices rising.

$$\varepsilon_{t-1} > 0 \text{ good news} \Rightarrow \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \quad \text{same old.}$$

$$\varepsilon_{t-1} < 0 \text{ bad news} \Rightarrow \sigma_t^2 = \alpha_0 + (\alpha_1 + \lambda) \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$a = 0.1$ to $a = 0.6$
graph

$\lambda > 0 \Rightarrow$ leverage effect.

GJR-GARCH Model

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \lambda(I_{t-1} \varepsilon_{t-1}^2) + \beta_1 \sigma_{t-1}^2$$

Depending on whether ε_{t-1} is above or below the threshold value of zero, ε_{t-1}^2 has different effects on the conditional variance σ_t^2 .

When ε_{t-1} is positive, the total effect is given by $\alpha_1 \varepsilon_{t-1}^2$;

When ε_{t-1} is negative, the total effect is given by $(\alpha_1 + \lambda) \varepsilon_{t-1}^2$.

So one would expect λ to be positive for bad news to have larger impacts.

①

```
spec_gjr11 <- ugarchspec(  
  variance.model = list(  
    model = "gjrGARCH",  
    garchorder = c(1, 1)  
  ),  
  mean.model = list(  
    armaorder = c(0, 0),  
    include.mean = TRUE  
  ),  
  distribution.model = "norm"  
)
```

<-| GJR/TGARCH
(ARCH(1), GARCH(1))

just an intercept

②

```
# Estimate GJR  
fit_gjr11 <- ugarchfit(  
  spec = spec_gjr11,  
  data = log_ret  
)
```

$\hat{\mu}$
 $\hat{\alpha}_0$

```

> fit_gjr11@fit$matcoef
      Estimate Std. Error  t value    Pr(>|t|)
mu      0.03335229 0.010417029   3.201708 1.366152e-03
omega   0.01108922 0.001930469   5.744313 9.229517e-09
alpha1  0.01301671 0.007591970   1.714536 8.643027e-02
beta1   0.90661144 0.009563280  94.801305 0.000000e+00
gamma1  0.12535877 0.014597083   8.587933 0.000000e+00
  
```

wow! $\rightarrow \lambda > 0 \Rightarrow$ evidence of leverage effect.

In the rugarch R-output, the coefficient gamma1 corresponds to the leverage parameter λ in our notation.

$\downarrow$
 λ

$H_0: \lambda = 0 \Rightarrow$ firmly rejected $\Rightarrow$ 0 p-value.

GJR-GARCH model

12:58

1:03 back.

The fitted GJR-GARCH model

$$r_t = 0.033 + \hat{\varepsilon}_t$$

$$\hat{\sigma}_t^2 = 0.011 + 0.013\hat{\varepsilon}_{t-1}^2 + \underbrace{0.125I_{t-1}}_{\hat{\lambda} \rightarrow \text{gamma1} \geq 0} \hat{\varepsilon}_{t-1}^2 + 0.907\hat{\sigma}_{t-1}^2$$

> 0 ≥ 0 wow! ≥ 0

The estimated value of λ is positive ($\hat{\lambda} = 0.125$), indicating the presence of leverage effects, and is statistically significantly different from zero at the 5% significance level since its t-statistic is greater than 1.96 and the p-value is less than 0.05.

$\log(-)$ is not defined. Thus no need to check coefficients to be positive.

Daniel Nelson proposed the exponential GARCH (EGARCH) model:

$$\log(\sigma_t^2) = \omega + \alpha_1 u_{t-1} + \gamma_1 (|u_{t-1}| - E|u_{t-1}|) + \beta_1 \log(\sigma_{t-1}^2).$$

where

↓
focus

→ wow!

$u_{t-1} = \frac{\varepsilon_{t-1}}{\sigma_{t-1}}$ is the "standardised" shock.

$u_t \sim iid N(0,1)$

If $\alpha_1 < 0$, then a negative shock has a larger effect on $\log(\sigma_t^2)$ than a positive shock of the same magnitude. This captures the leverage effect.

shocks = ε_{t-1}

negative shock means $u_{t-1} < 0$

IF $\alpha_1 < 0 \Rightarrow \alpha_1 u_{t-1} > 0 \Rightarrow \text{increase } \log(\sigma_t^2) \Rightarrow \text{increases volatility} \Rightarrow \text{panic}$
↓
evidence of leverage effect.

EGARCH

plays no role for effect of leverage.

$$\log(\sigma_t^2) = \omega + \alpha_1 u_{t-1} + \gamma_1 (|u_{t-1}| - E|u_{t-1}|) + \beta_1 \log(\sigma_{t-1}^2).$$

If we assume $u_{t-1} \sim N(0, 1)$ then

$$E|u_{t-1}| = \sqrt{\frac{2}{\pi}} \approx 0.7979.$$

Another advantage of the EGARCH model over the basic GARCH model is that the conditional variance σ_t^2 is guaranteed to be positive regardless of the values of the coefficients in the model, because the logarithm of σ_t^2 instead of σ_t^2 itself is modelled.

Therefore, the conditional variance is automatically positive, and we do not need to impose the usual non-negativity restrictions on the coefficients.

great!

1

```
spec_egarch11 <- ugarchspec(
  variance.model = list(
    model = "eGARCH",
    garchorder = c(1, 1)
  ),
  mean.model = list(
    armaorder = c(0, 0),
    include.mean = TRUE
  ),
  distribution.model = "norm"
)
```

<-- EGARCH
c(arch, garch)

$E|u_{t-1}| = \sqrt{\frac{2}{\pi}}$

2

```
# Estimate E-GARCH
fit_egarch11 <- ugarchfit(
  spec = spec_egarch11,
  data = log_ret
)
```

```
> fit_egarch11@fit$matcoef
```

	Estimate	Std. Error	t value	Pr(> t)
mu	0.027972886	0.009270186	3.017511	0.0025485991
omega	-0.007788339	0.002110986	-3.689433	0.0002247547
alpha1	-0.101967927	0.007830779	-13.021428	0.0000000000
beta1	0.979526740	0.001050333	932.586652	0.0000000000
gamma1	0.143450793	0.012157376	11.799486	0.0000000000

$\hat{\alpha}_0$



wow!

no role

leverage-effect $\alpha_1 < 0$

EGARCH

In the R output, the estimated α_1 coefficient for the leverage effect is denoted by `alpha1` and is negative in this case. That is $\hat{\alpha}_1 = -0.102$.

The fitted EGARCH model:

$$r_t = 0.028 + \hat{\varepsilon}_t$$

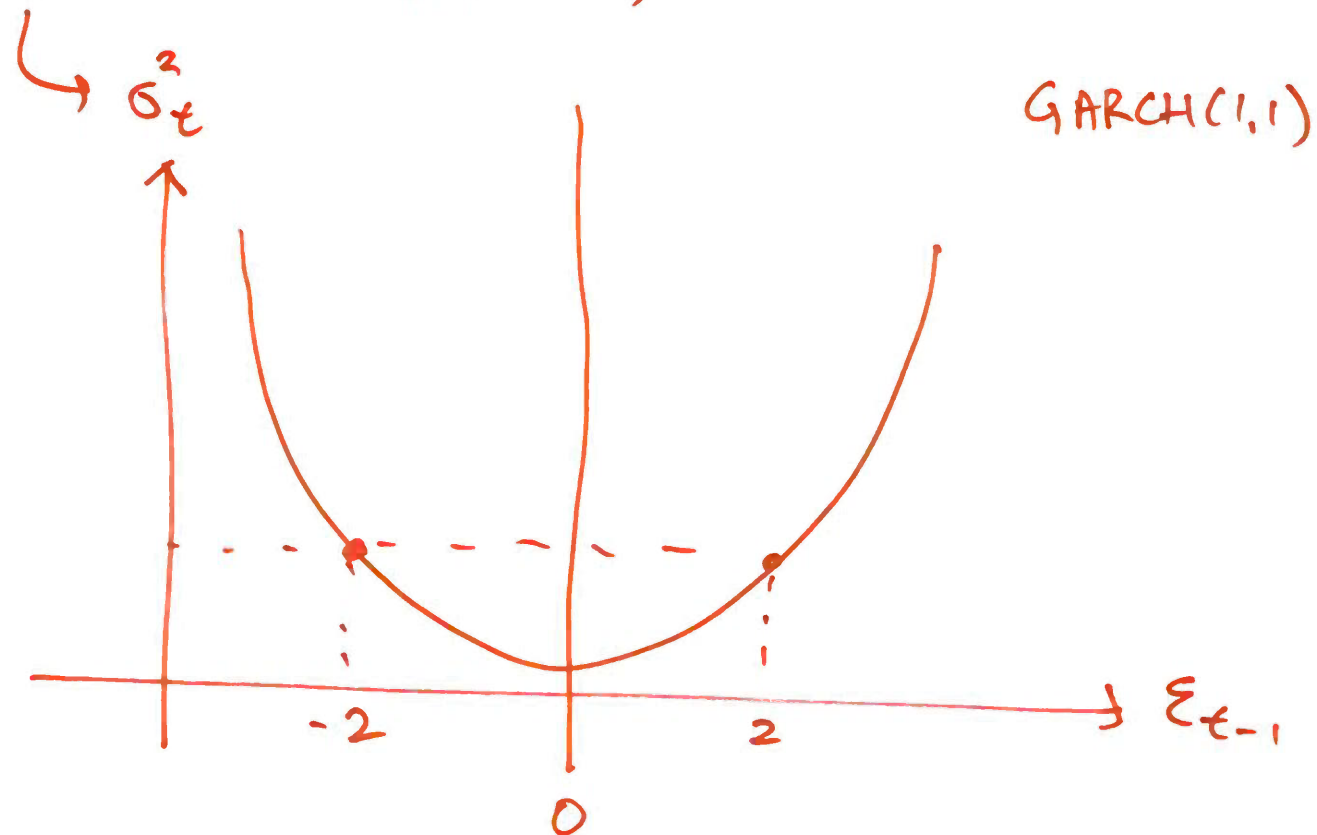
$$\begin{aligned} \log(\sigma_t^2) = & -0.0079 - \overset{\checkmark}{0.102}u_{t-1} + 0.1435 (|u_{t-1}| - E|u_{t-1}|) \\ & + \underline{0.9795} \log(\sigma_{t-1}^2). \\ & \text{wow!} \end{aligned}$$

The t -statistic for testing $\alpha_1 = 0$ is -13.021 with a p-value approximately equal to zero.

Points about asymmetric GARCH models

- ▶ Forecasting one-step-ahead with these models is *is good* virtually the same as with an ARCH or GARCH model.
- ▶ Forecasting more than one period ahead with asymmetric GARCH models is not straightforward and is usually done using simulation techniques.

News Impact Curves $NIC(x)$



News Impact Curve

"shocks"



The 'news impact curve (NIC)' is a plot of σ_t^2 against ε_{t-1} .

Consider an ARCH(1) model

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2$$

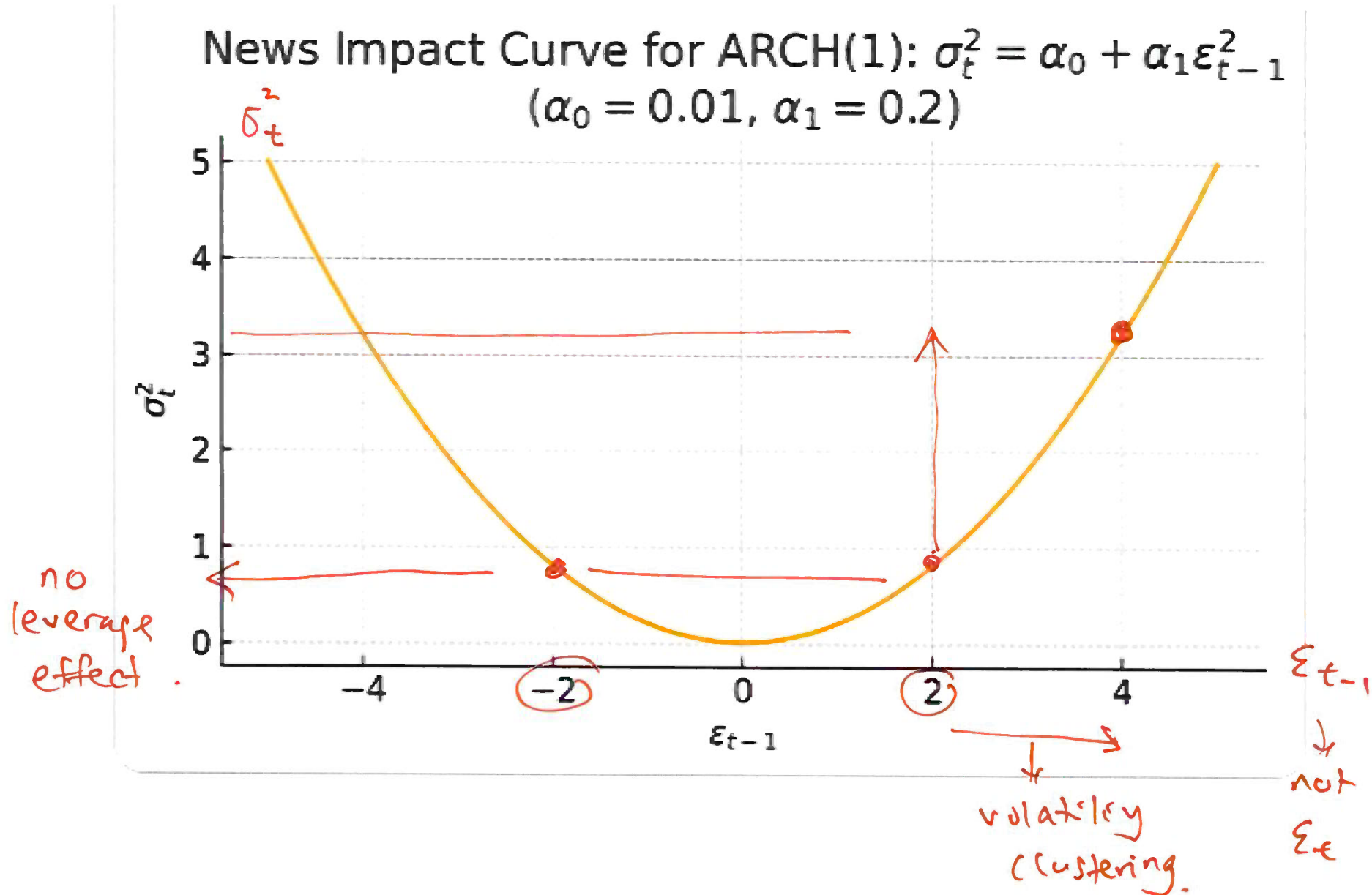
||
 ≈ 0

$\alpha_1 = 0.3$

Interpretation

- ▶ "Small" shocks $\varepsilon_{t-1} \simeq 0.0$ result in a small variance $\sigma_t^2 \simeq \alpha_0$.
- ▶ "Large" shocks, positive or negative, result in an increase in the variance in the next period if $\alpha_1 > 0$.

The NIC plots the next-period conditional variance, σ_t^2 , against the size of yesterday's shock, ε_{t-1} .



Because only the square of the shock enters, the curve is simply the upward-opening parabola.

Symmetry: positive and negative shocks of the same magnitude have identical effects.

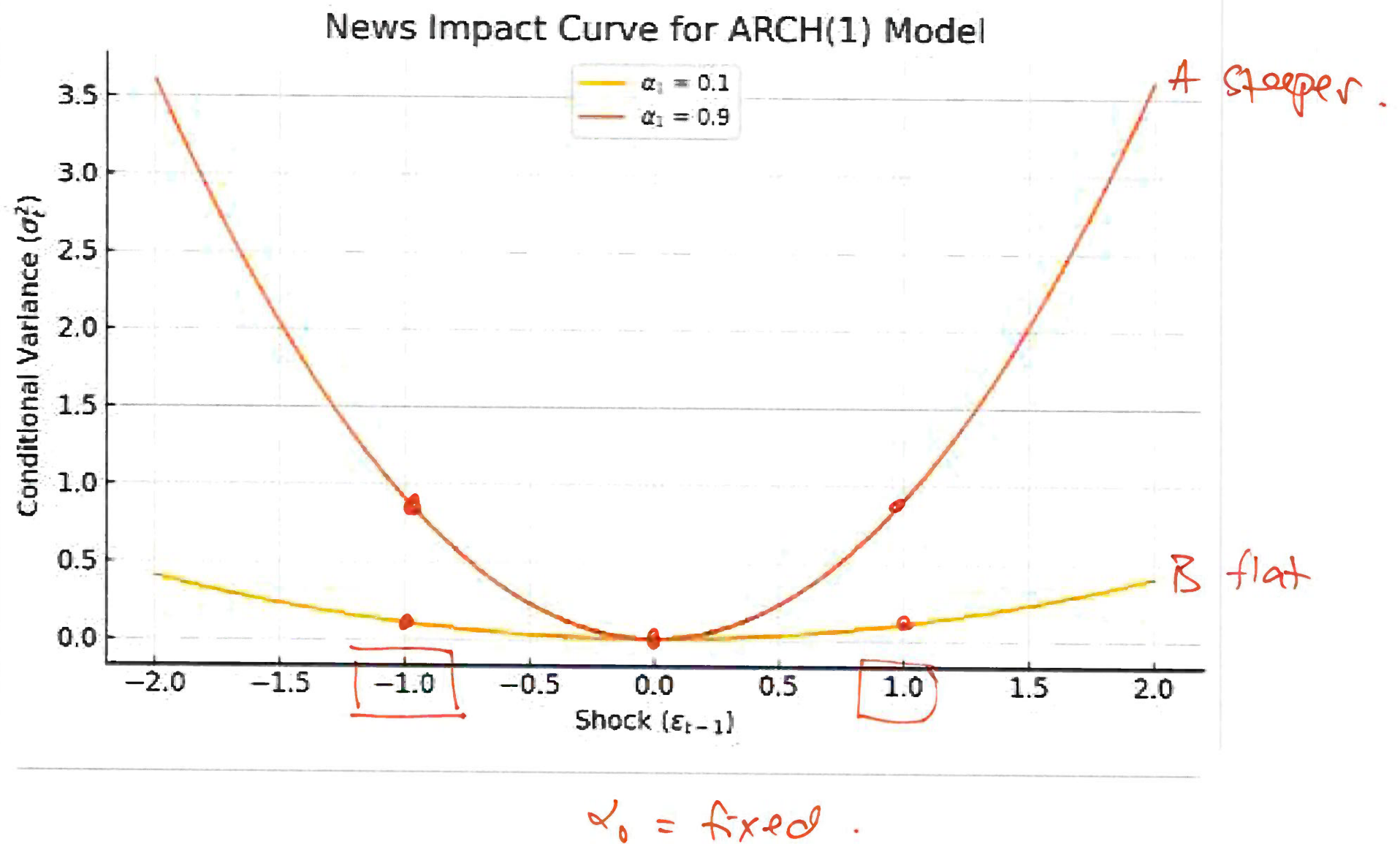
Convexity: the impact grows quadratically; extreme shocks inflate volatility much faster than small shocks.

Intercept: α_0 is the “baseline” variance when $\varepsilon_{t-1} = 0$ which is when the news is neutral.

Slope/curvature: α_1 governs how sensitive volatility is to news; higher α_1 steepens the parabola.

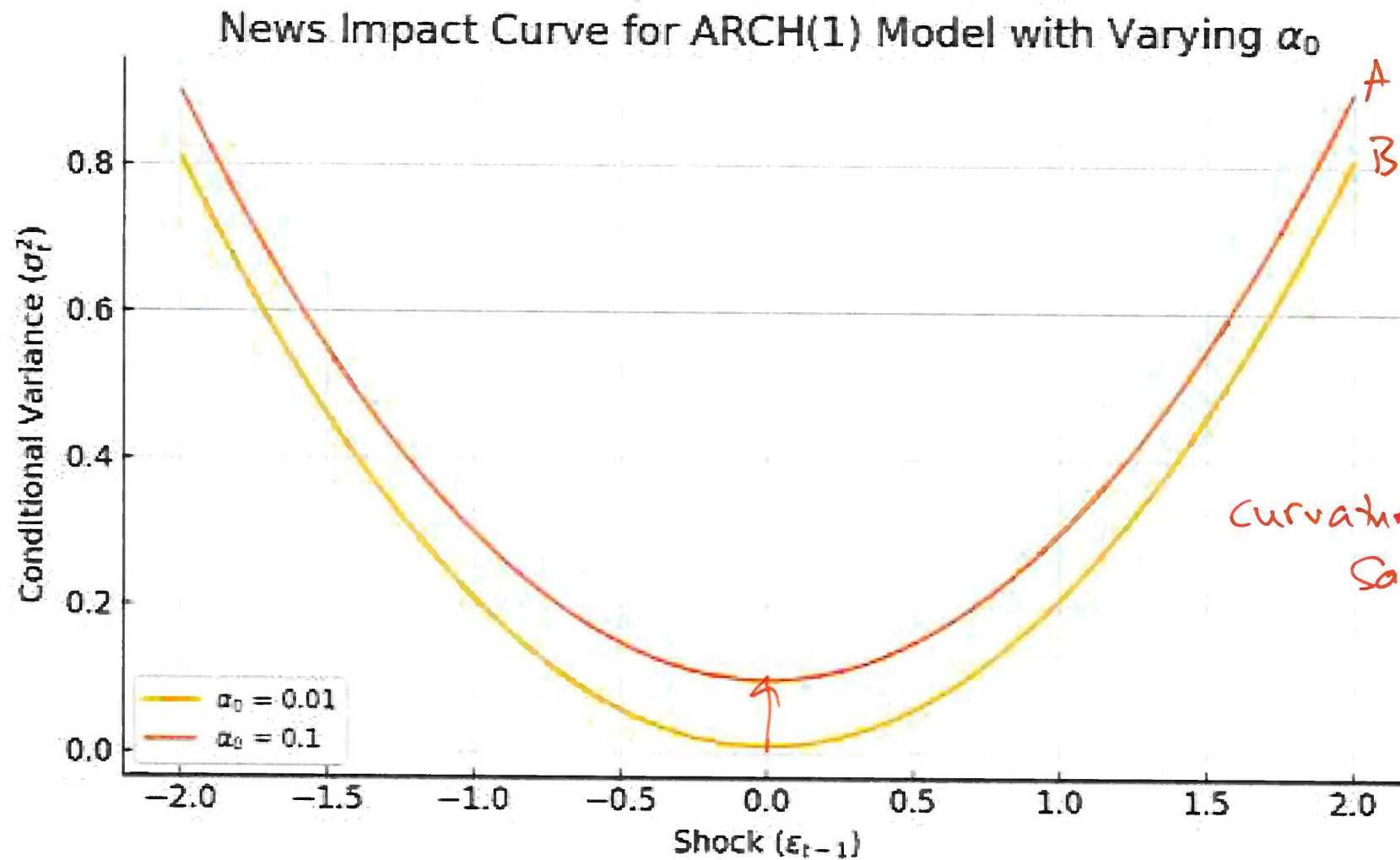
No leverage effect: Because the curve is symmetric, ARCH(1) cannot capture the common finding that bad news (negative returns) increases volatility more than good news of identical size.

Volatility clustering: Large ε_{t-1}^2 pushes σ_t^2 up, which in turn makes large future shocks more likely, creating the familiar ARCH/GARCH volatility-clustering pattern.



For a fixed $\alpha_0 = 0.01$, the plot in the previous slide shows how the conditional variance σ_t^2 responds to shocks ε_{t-1} under two ARCH(1) specifications:

- ▶ With $\alpha_1 = 0.1$, the response is relatively flat.
- ▶ With $\alpha_1 = 0.9$, the response is much steeper, indicating a stronger sensitivity of volatility to shocks.

α_1 fixed.

NIC

For a fixed $\alpha_1 = 0.2$, the plot in the previous slide illustrates how increasing α_0 from 0.01 to 0.1 in an ARCH(1) model raises the baseline level of conditional variance σ_t^2 , while the curvature (i.e. sensitivity to shocks ε_{t-1}) remains unchanged.

α_0 determines the **vertical position** of the NIC.



w my notes.

↓
shifting curve up / down.

$$\text{GARCH}(1,1): \sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$\underbrace{\hspace{10em}}$
 $\hookrightarrow$ R variance.

For a GARCH(1,1) model, all lagged conditional variances are evaluated at the level of unconditional variance in the NIC.

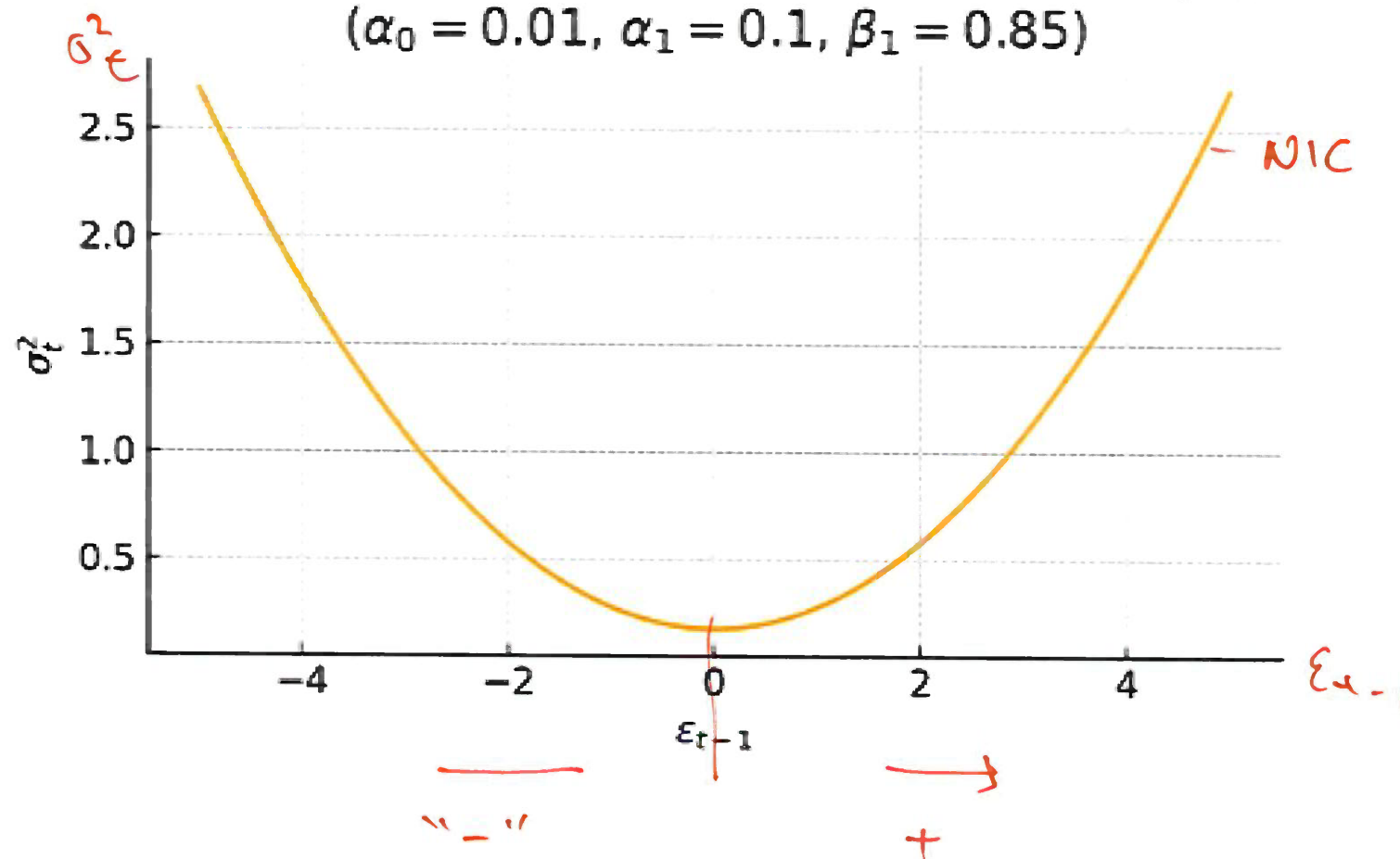
The NIC is given by

$$\sigma_t^2 = \alpha_0 + \beta_1 \underbrace{\frac{\alpha_0}{1 - \alpha_1 - \beta_1}}_{\hookrightarrow \text{R Variance}} + \alpha_1 \varepsilon_{t-1}^2$$

ω

The plot is symmetric around $\varepsilon_{t-1} = 0$. The steepness of the plot depends on α_1 .

News Impact Curve for GARCH(1,1): $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$
 ($\alpha_0 = 0.01$, $\alpha_1 = 0.1$, $\beta_1 = 0.85$)



GJR-GARCH

For GJR-GARCH:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \overset{\text{new}}{\lambda(I_{t-1} \varepsilon_{t-1}^2)} + \beta_1 \underbrace{\sigma_{t-1}^2}_{\text{with LR}}$$

The unconditional variance is

$$\mathbb{V}(\varepsilon_t) = \frac{\alpha_0}{1 - \alpha_1 - \frac{1}{2}\lambda - \beta_1}$$

This is because under a symmetric distribution (e.g. standard normal), negative shocks occur half the time. Therefore,

$$E[I_{t-1} \varepsilon_{t-1}^2] = \frac{1}{2} \mathbb{V}(\varepsilon_t).$$

Recall

$$I_{t-1} = \begin{cases} 1 & \varepsilon_{t-1} \leq 0 \\ 0 & \text{else} \end{cases}$$

News impact curves for GJR-GARCH model

For GJR-GARCH: $\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \lambda(I_{t-1} \varepsilon_{t-1}^2) + \beta_1 \sigma_{t-1}^2$

The NIC is

w my note.

$$\sigma_t^2 = \begin{cases} A + \alpha_1 \varepsilon_{t-1}^2 & \text{for } \varepsilon_{t-1} > 0 \quad \text{no body care} \\ A + (\alpha_1 + \lambda) \varepsilon_{t-1}^2 & \text{else} \quad \text{negative news} \Rightarrow \boxed{\lambda > 0 \text{ must}} \end{cases}$$

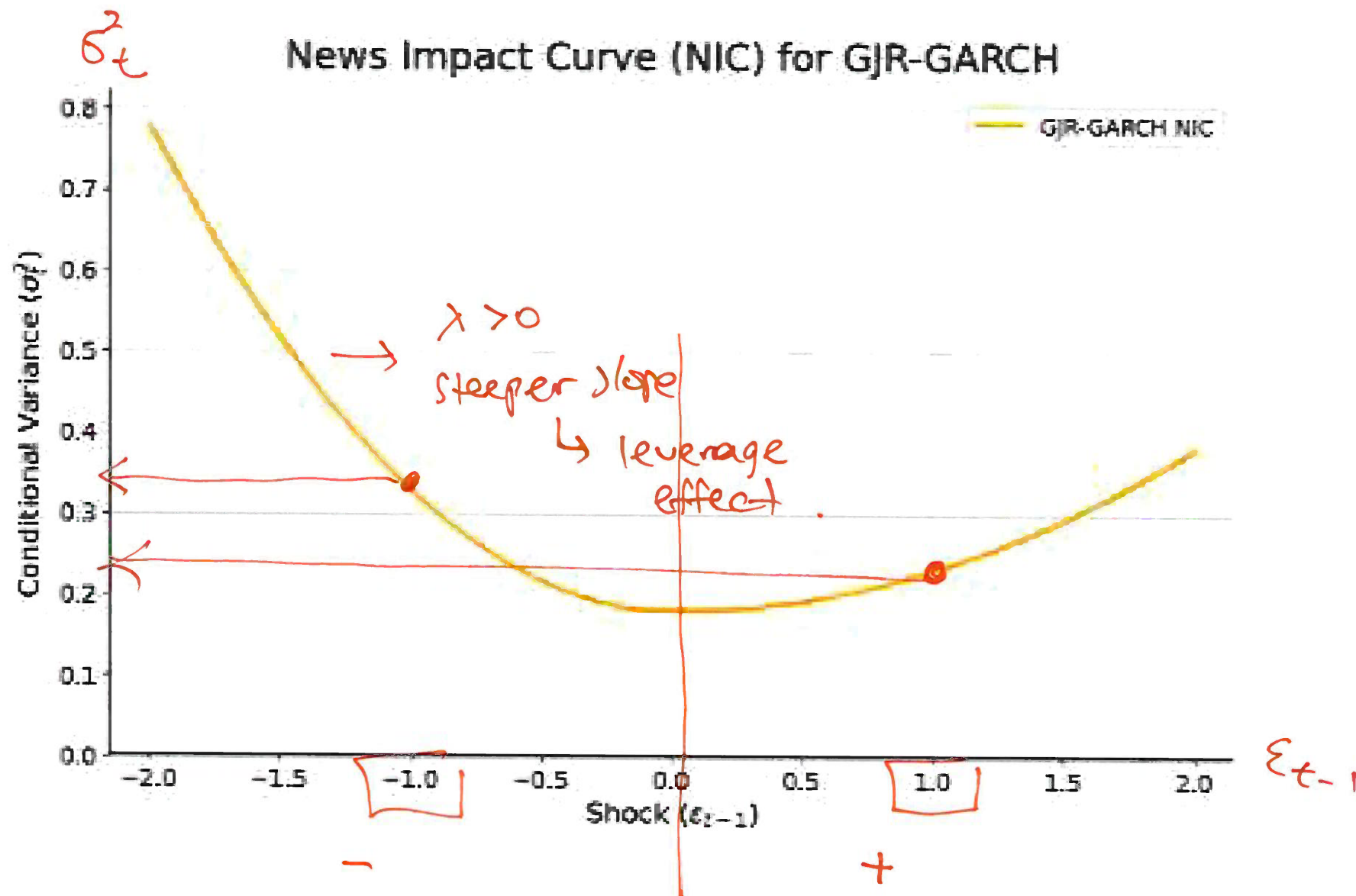
$$A = \alpha_0 + \beta_1 \frac{\alpha_0}{1 - (\alpha_1 + \lambda/2) - \beta_1}$$

LR variance.

⇒ a number
base case

Shift curve up /
down

In the GJR-GARCH model the curve has its minimum at $\varepsilon_{t-1} = 0$.



GJR-GARCH

The News Impact Curve (NIC) shown here is for a GJR-GARCH model with $\alpha_0 = 0.01$, $\alpha_1 = 0.05$, $\lambda = 0.10$ and $\beta_1 = 0.85$

It shows how negative shocks increase conditional variance more than positive shocks of the same magnitude due to the leverage effect $\lambda > 0$.

Robert F. Engle shared the 2003 Nobel Prize in Economics for methods of analysing economic time series with time-varying volatility, especially ARCH models. GARCH was later introduced by Tim Bollerslev as a generalisation of Engle's ARCH model.

